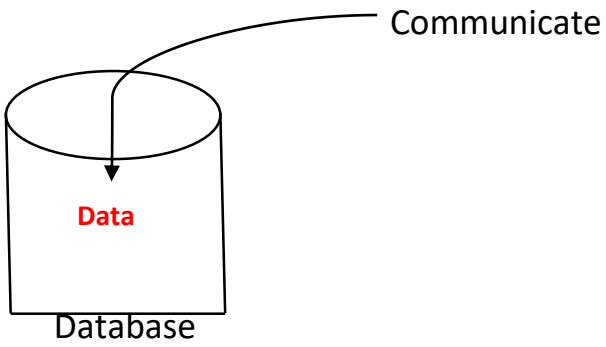
Introduction of SQL

SQL (Structured Query Language): It is a language which is used to interact/communicate with database.



History of SQL

SQL ------> 1970----> Raymond Boyce

(Data Scientist) IBM

When he invented SQL. He named SQL as SEQUEL (**Simple English Query Language**)

Then after in:

1986 ---> ANSI (American National Standard Institute)

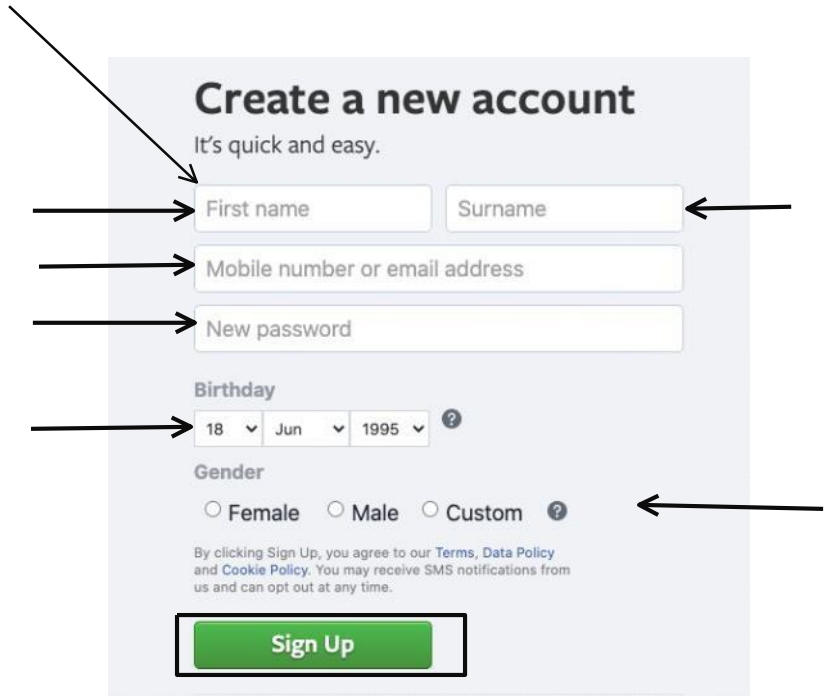
ANSI took over SEQUEL and changed its name to SQL.

DATA

"Data is a raw fact which describes attributes of an entity."

- Properties are also called attributes.
- Entity is also called an object.

Properties or Attributes



EXAMPLE:

Entity/Object

Water Bottle

Attribute/Property

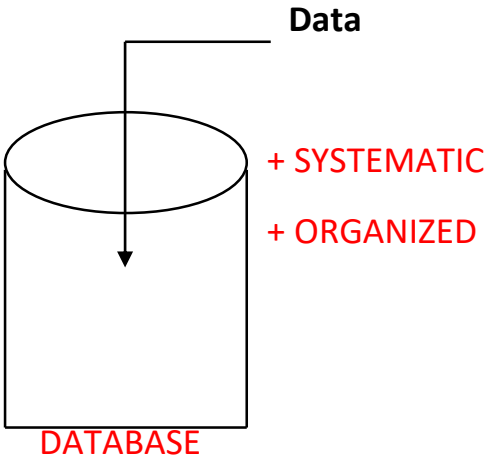
Height: **20cms**

Color: **Green**

Capacity: **500 ml**

DATABASE

"Database is a place or a medium in which we store the data in systematic & organized manner."



- The basic operations that can be performed on a database are

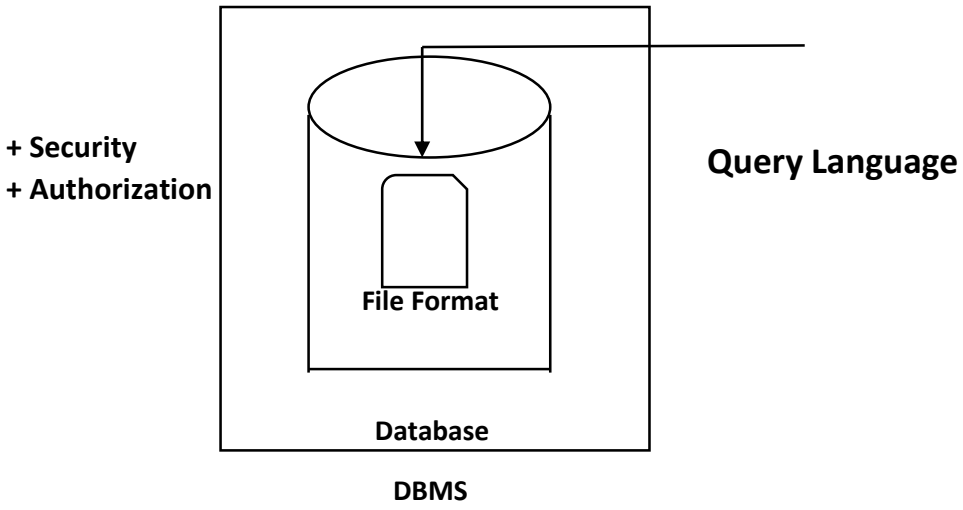
  - CREATE/INSERT
  - READ/RETRIEVE
  - UPDATE/MODIFY
  - DELETE/DROP

- These operations are referred to as **"CRUD"** operations.

DBMS (Database Management System)

"It is a software which is used to maintain and manage the database"

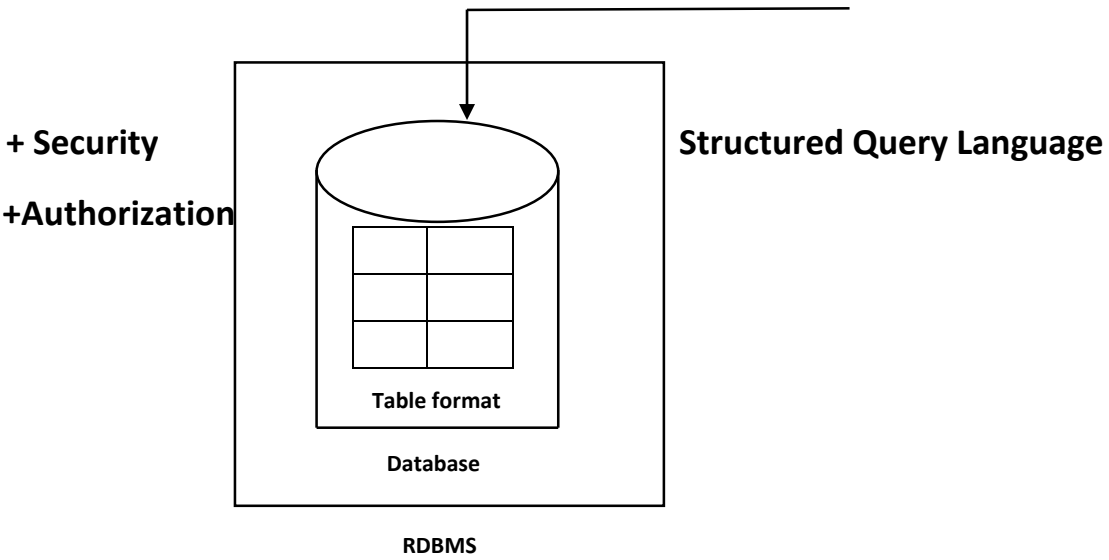
- In DBMS software we have two main features
  - Security
  - Authorization
- In DBMS we store the data in file format.
- To communicate with DBMS, we are using Query Language.



RDBMS (Relational Database Management System)

"It is a type of DBMS software used to maintain & manage the database."

- In RDBMS software we have two main features
  - Security
  - Authorization
- In RDBMS we store the data in the form of table (rows & columns)
- To communicate with RDBMS, here we are using Structured Query Language.



Difference Between DBMS & RDBMS

DBMS	RDBMS
➢ It is a software used to maintain and manage the database.	It is type of DBMS software used to maintain and manage the database
➢ In DBMS we store data in File format.	In RDBMS we store data in Table format.
➢ To communicate with DBMS, we use query language.	To communicate with RDBMS we are using SQL
➢ Performing CRUD operation is Difficult compared to RDBMS	Performing CRUD operation is easy.

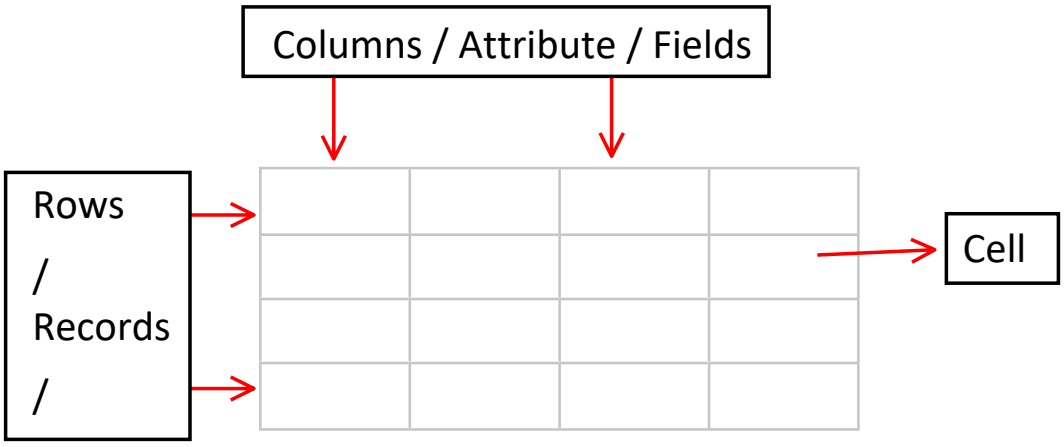
RELATIONAL MODEL

- Relational model was designed by famous data scientist **E.F CODD (Edgar Frank codd)**
- In Relational model we store data in the form of rows & columns**(Tables).**
- Any DBMS which follows Relational model becomes RDBMS.

- Any DBMS which follows rules of E.F CODD becomes RDBMS.

TABLE

"It is a logical organization of data which consists of columns & rows."



Columns: It describes single properties of all the entity
Rows: It describes all the properties of a single entity
Cells: It is a smallest unit of a table **OR** intersection between rows & columns generates a space that is called as cell.

EXAMPLE:

E_ID	Ename	Sal
1	SMITH	1000
2	ALLEN	1500
3	CLARK	2000

RULES OF E.F CODD

1. **The data entered into a cell must always be a single valued data.**

   Example:

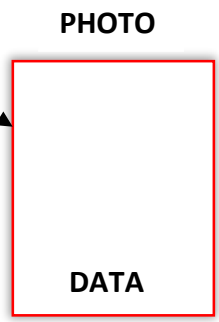
EID	ENAME	PH_NO
1	SMITH	1234567890
2	ALLEN	1234568790, 4567889012
3	CLARK	1234689060

EID	ENAME	PH_NO	ALTERNATE_NO
1	SMITH	1234567890	
2	ALLEN	1234568790	4567889012
3	CLARK	1234689060	

2. **In RDBMS, we can store everything in the form of table including metadata.**

Metadata: The details about data or data about data is known as **metadata**.

EID	ENAME	PHOTO
1	SMITH	
2	ALLEN	
3	CLARK	



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Meta Table

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Meta Table: The table which stores meta data is called as meta table.

3. **According to E.F CODD, we can store data in multiple tables if needed we can establish connection between the tables using key attributes.**

Need to establish connection

Employee

E_id	Ename	Age
101	Subhan	30
102	Priya	28
103	Aditya	32

Department

D_no	Dname	loc
10	D1	L1
30	D2	L2
20	D3	L3

4. The data entered into a table can be verified in two steps
   - By assigning Datatypes
   - By assigning Constraints